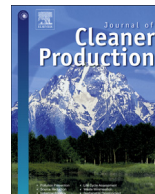


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Multi criteria decision making approaches for green supplier evaluation and selection: a literature review

Kannan Govindan^{a,*}, Sivakumar Rajendran^b, Joseph Sarkis^c, P. Murugesan^d^a Operations and Supply Chain Management, Department of Business and Economics, University of Southern Denmark, Odense 5230, Denmark^b Department of Mechanical Engineering, Nadar Saraswathi College of Engineering and Technology, Theni 625531, India^c Department of Management, School of Business, Worcester Polytechnic Institute, 100 Institute Road, Worcester MA 01609-2280, USA^d PVP College of Engineering & Technology for Women, K. Singarakottai, Dindigul 624708, Tamil Nadu, India

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ABSTRACT

A large and growing body of literature to supplier evaluation and selection exists. Literature on green supplier evaluation that considers environmental factors are relatively limited. Recently, in supply chain management decision making, approaches for evaluating green supplier performance have used both qualitative and quantitative environmental data. Given this evolving research field, the goal and purpose of this paper is to analyze research in international scientific journals and international conference proceedings that focus on green supplier selection. We propose the following questions that will be answered: (i) which selection approaches are commonly applied?, (ii) what environmental and other selection criteria for green supplier management are popular?, (iii) and what limitations exist? Published research from 1997 to 2011 is structurally reviewed based on the first two questions. We find that the applied techniques are mostly fuzzy based single model approaches. The most common criterion considered for green supplier selection was “environmental management systems”. A further critical analysis is completed and gaps in the current literature are identified. These gaps help us to identify improvements for green supplier selection process and possible future directions.

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1. Introduction

Supplier selection is a key operational task for developing sustainable supply chain partnerships. Environmental, social, and economic dimensions must all be considered in order to select a well-rounded sustainable supplier, one that can enhance supply chain performance. Part of the supplier selection process involves supplier evaluation together with selection, which is an important issue to supply chain and production and operation management literature (Motwani and Youssef, 1999).

Currently, due to outsourcing initiatives, organizations have become more dependent on suppliers making it more critical to choose and evaluate their supplier performance. Supplier evaluation and selection requires the consideration of multiple objectives and criteria (Bhutta and Huq, 2002). Research has been robust in this field with study including adopting approaches and implementation from a wide range of mathematical practices and

methodologies. Consequently, numerous multi-criteria decision support tools have been developed for structuring and supporting such decisions (Wu et al., 2010).

Typically, when organizations seek to develop or choose a supplier evaluation and selection method, the organization's specific requirements are introduced. Therefore, model flexibility and a choice from a range of different selection methods with different applications are needed.

One such topic of increasing interest is the green, environmental supply chain management issues. This area involves and screens suppliers based on their organizational environmental performance, and distinguishes whether a certain supplier's performance is baseline (e.g. meeting regulatory requirements) or advanced (e.g. collaborative green product design) practice (Rao, 2002). Traditionally, environmental protection issues were considered in the production process along with traditional supplier selection criteria (Cao, 2011). Due to this current environmental awareness, suppliers have adopted many green policies and practices.

Recent established research has utilized supplier selection mechanisms which consider environmental protection issues (Humphreys et al., 2003a, b; Yeh and Chuang, 2011). A potentially effective way of managing a company's environmental policy is by

* Corresponding author. Tel.: +45 6550 3188, +45 40469925 (mobile); fax: +45 6550 3237.

E-mail address: gov@sam.sdu.dk (K. Govindan).

linking it closely with purchasing function activities, i.e. through supplier selection (Humphreys et al., 2003a, b). Provisional guidelines for environmentally evaluating the suppliers have existed for some time (Lamming and Hampson, 1996). In support of these early efforts, environmental performance measurement frameworks were also introduced (Noci, 1997; Hervani et al., 2005). Noci (1997) identified performance criteria to consider in the green supplier selection process and suggested techniques for effective supplier selection from an environmental viewpoint. Shen et al. (2013) proposed a fuzzy approach for evaluating the green suppliers. Also, Hervani et al. (2005) provided an integrative framework which focuses on study, design, and evaluation aspects of green supply chain management performance tools.

Here, we carefully examine green supplier selection papers that appeared from 1996 to 2011 in international scientific journals available in electronic databases such as Elsevier's Science Direct, Emerald Publishers, Springer, IEEE, Taylor & Francis, and Google Scholar. This research analyzes the papers published in international conference proceedings and summarizes the contributions made by researchers on green supplier selection at the conferences. In our search of the databases, we used the following keywords: green supplier evaluation, green supplier selection, green supply chain management, green criteria, and supplier selection. In our examination, we highlight the methodologies and environmental criteria utilized. As our first major objective, we analyze multi-criteria decision-making approaches such as the analytical hierarchy process (AHP), analytical network process (ANP), case based reasoning (CBR), data envelopment analysis (DEA), fuzzy set theory, genetic algorithm (GA), mathematical programming and their hybrids. The second objective is an analysis of the green criteria used, including issues such as green products, green knowledge transfer, green image, recycling, environmental management systems, environmental certificates, and other criteria utilized in green supplier selection models. This information is useful to researchers and practitioners by clearly pointing to applicable tools and content (performance measures) that have and could be applied to environmental supplier selection. A critical analysis will be helpful in guiding future research and application, and in clearly identifying the capabilities, boundaries, and limitations of the tools.

To achieve these goals we begin in Section 2 by giving an overview of research and literature on supplier selection and green supplier selection. In Section 3 we describe the research methodology, followed by the classification method presented in Section 4. Section 4 also summarizes the literature based on multi-criteria decision making approaches and environmental criteria for the selection process. Section 5 presents an observation by the discussion about the most prevalently used approaches and most popular evaluation criteria considered for the selection process. Section 6 discusses the results and future work, and Section 7 focuses on drawing final conclusions.

2. Supplier selection and green supplier selection

Research in supplier selection has gained particular importance over the past few decades as organizations started to focus on issues of core competence and to outsource less profitable activities to supply chain partners (Arnold, 2000; Lonsdale and Cox, 2000; Quinn, 1999). This evolution included a more strategic focus on the buyer–supplier relationship where close collaboration was necessary and certain skills and capabilities were required (Sarkis and Talluri, 2002).

These skills and capabilities required greater scrutiny by buyer organizations, which led to the more complex tasks of identifying and selecting strategic supply chain partners because some suppliers have limited capacity or other constraints (Kannan et al.,

2013). This real world complexity in the outsourcing and vendor selection process generated the need to help organizations make more thoughtful and simplified decisions. Simplifying complex managerial decision making is the role of many pragmatic theories and models (Williamson, 2008).

Models for supplier selection represent only one of over a dozen supply chain management areas (see Badole et al., 2013 for a comprehensive review of supply chain modeling literature). Thus, it is easy to see that a strategic direction in supplier management practices requires the ability to take multiple criteria and measures in order to arrive at a clear and straightforward prioritization or final selection (Ho et al., 2010). The extensive nature and modeling complexity of the regular supplier selection process makes the problem heavily reliant on multiple criteria decision models.

Multiple criteria decision making is not only for the purposes of simplification and arriving at a clear decision, but it also allows researchers and management to balance a variety of criteria, many of them conflicting (Sarkis and Talluri, 2002; Chai et al., 2013). This level of complexity and tradeoffs is even more evident when it includes the additional dimensions of greening of supply chains (Bai and Sarkis, 2010b).

Environmental and economic tradeoffs are commonplace in organizational decision making, whether it is based on supplier selection or other management decisions such as technology and product selection (Ajukumar and Gandhi, 2013; Khalili and Duecker, 2013; Sarkis, 2003; Sarkis et al., 2012). The tradeoffs are typically based on a variety of operational and strategic sustainability metrics that need careful integration, with or without management input.

The supplier selection process has only recently (within the last decade) started to integrate various environmental dimensions. The decision models will necessarily become more complex due to the many new dimensions brought in by green supply chain efforts, where the tradeoffs become more evident and numerous. The decisions will also include more intangible dimensions such as reputation, supply chain risk, business continuity, and social impact. These new criteria and dimensions required rethinking some of the more established approaches and models. In addition, decision makers, or agents that influence the decisions, continue to grow when environmental factors come into play (Herva and Roca, 2013). Extended stakeholder influence, input, and considerations all start to play a larger role in green supply chain management (Govindan et al., 2013a; Sarkis et al., 2012; Bai and Sarkis, 2010a). Whether these issues are captured in the evolving models is something to consider as we provide the review. Before reviewing and critically examining the growing literature in green supplier selection, we introduce the methodology we partake to select the research we seek to review.

3. Research methodology

Literature reviews are valuable comprehensive studies used to investigate research in emergent fields and to help guide future research and directions (Lage Junior and Godinho Filho, 2010; Govindan, 2013; Govindan et al., 2013b). This paper presents the following research methodology:

- Step 1: Perform a literature review regarding green supplier evaluation and selection.
- Step 2: Develop a classification framework using criteria.
- Step 3: Segregate and tabulate the literatures based on the framework
- Step 4: Present the literature review using the classification framework to organize the review.

Step 5: Analyze the review and present suggestions for future work.

To accomplish this study's objectives, this research examines journals that incorporate a number of green supply chain management studies. The literature review is a usual method to investigate thoroughly different approaches of the topic to be studied (Lage Junior and Godinho Filho, 2010). Peer-reviewed academic journals and proceedings are most commonly used to acquire information and report new research findings (Ngai et al., 2008; Lage Junior and Godinho Filho, 2010). Thirty-three papers appearing from 1996 to 2011 are targeted using various publisher and publicly available electronic databases, including 13 papers of Elsevier's Science Direct (39.39%), 3 papers of Emerald Publishers (9.09%), 2 papers of Springer (6.06%), 12 papers of IEEE (36.36%), 3 papers of Taylor & Francis (9.09%) publications, and the Google Scholar database.

The following keywords were used in some form in the search databases: green supplier evaluation, green supplier selection, green supply chain management, green criteria, and supplier selection.

4. A classification

After collecting the literature, a classification framework (Fig. 1) is constructed. This research focuses only on green supplier selection in which environmental criteria is taken into account. Hence, the traditional criterion has not been considered in this research. The classification is based on two main categories:

- a) Decision Making Methodology Base
 - (i) Individual Methodology Approach
 - (ii) Integrated Methodology Approach
- b) Criteria Selection Base
 - (i) Environmental Criteria

The remainder of this section focuses on each of the various categories identified in this classification framework.

4.1. Multi-criteria decision-making – individual methodology approach

Researchers have developed many tools for multiple criteria decision making, and many of the latest integrate fuzzy logic. From the analysis of articles, the research has applied multi-criteria decision-making approaches in both individual and integrated ways. A summary of the multi-criteria decision-making approaches, using an authorship ordering, applied to green supplier selection is given in Appendix 1.

4.1.1. Analytical hierarchy process (AHP)

Six out of thirty six (16.6%) journal articles we identified propose AHP (including fuzzy AHP, FEHP) for green supplier selection

process. The practical applications and evaluating criteria used in these approaches are summarized in Appendix 2.

Noci (1997) proposed an AHP based method for assessing a vendor's environmental efficiency. The tool was applied in an automotive case using five steps to calculate suppliers' capabilities to achieve high environmental performance; suppliers who achieved the highest scores were preferable and recommended for selection.

Handfield et al. (2002) illustrated the case of AHP as a decision support tool to help managers understand the trade-offs between environmental dimensions. They demonstrated how AHP can be used to evaluate the relative importance of various environmental traits and to access the relative performance of several suppliers along with the traits.

Lu et al. (2007) applied AHP to evaluate and coordinate green suppliers in a project environment. A weighting system using a fuzzy logic process is used to modify AHP.

Chiou et al. (2008) also adjusted AHP and applied FAHP for selection of green suppliers using 6 criteria with 24 sub criteria. The tool application was to determine the relative importance of selecting green suppliers across a multicultural setting including American, Japanese, and Taiwanese electronic industries in China.

Lee et al. (2009) applied FAHP integrated with the Delphi method for green supplier evaluation. The Delphi method was initially used to differentiate the criteria for evaluating traditional and green suppliers. FAHP is used to solve the green supplier selection process; they focused on the efficiencies of FAHP. They used 11 main criteria and 41 sub criteria.

Grisi et al. (2010) implemented a fuzzy AHP for green supplier evaluation using a 7 step approach. Fuzzy logic was adopted to overcome uncertainty arising from human qualitative judgment.

4.1.2. The analytic network process (ANP)

Four papers (11.1%) in our data set proposed ANP to handle the green supplier evaluation and selection process. The illustrative application and evaluation criteria in these papers are summarized in Appendix 3.

Hsu and Hu (2007, 2009) applied ANP for green supplier selection to further incorporate interdependencies among decision structure components. Their argument was that ANP captures both quantitative and qualitative criteria, reflecting a more realistic result offering managerial insights while selecting suppliers systematically.

Buyukozkan and Cifci (2010, 2011) developed a novel approach based on a fuzzy ANP model within a multi-person decision making scheme under incomplete preference relationships. This method's advantages allows for sufficient evaluation by using the provided preference information and maintaining the evaluation consistency.

4.1.3. Mathematical programming

One article in our data set (2.77%) proposes mathematical programming for the green supplier selection process. The application and evaluating criteria used in the approaches are summarized in Appendix 4.

Yeh and Chuang (2011) developed an optimal mathematical planning model for green partner selection which involves four objectives. They adopted two multi-objective genetic algorithms (MOGA) to find the set of Pareto-solution, which is utilized the weighted sum approach and compared the average number of Pareto operational solution and CPU times of the two algorithms.

4.1.4. Other approaches

Fourteen papers (38.8%) in the data set propose other approaches, and their application setting and evaluating criteria are summarized in Appendix 5.

Zhang et al., (2003) proposed a fuzzy multi-agent decision making strategy to facilitate supplier management. They evaluate

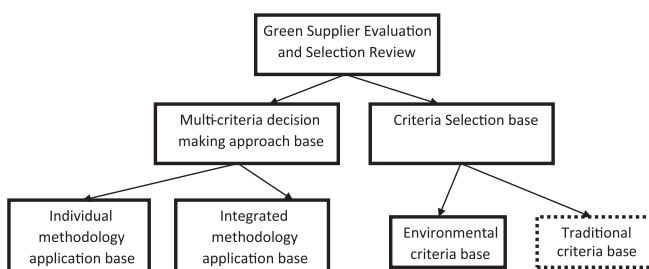


Fig. 1. Classification framework for green supplier evaluation and selection review.

the environmental performance of the suppliers and the life cycle environmental impact of a purchased product.

Vachon and Klassen (2006) applied Chi-Square Test for exploring the operational performance of green partnership in the supply chain. In this study, the data from the survey was used for testing the linkage between the green project partnership and five performance indicators.

Humphreys et al. (2006) proposed the fuzzy based system for green supplier selection based on the quantitative and qualitative environmental criteria. The major benefit of this system is its computational parsimony. The proposed system is capable of implementing a range of user priorities that influences to varying degrees the system output. The hierarchical fuzzy system, with scalable fuzzy membership functions employed, imparts user priorities into the system that can weakly or strongly influence the supplier selection process.

Yang and Wu (2007) constructed a multi-level grey entropy synthetic evaluation model for evaluation of a green supplier. The evaluation process indicates that it was easier to implement the method in enterprises. The main aspect of this model was that fewer hierarchy weight factors were needed. Arguably, their result is more objective than other evaluation methods.

Yang and Wu (2008) developed a multi-level extensible synthetic evaluation model based on the grey entropy synthetic evaluation model for green supplier evaluation. This model overcomes the lower weight factor avoidance and the result was more objective than the other evaluation methods.

Bai and Sarkis (2010a, b) applied Rough set methodology which utilizes an incomplete information approach which is more realistic in some data poor environments for green supplier development. There is also a limitation and practical managerial concerns with this multi-criteria decision-making approach in which the number of rules can become quite large depending on the volume of the data which is to be analyzed.

Kumar and Jain (2010) proposed a comprehensive approach DEA for green supplier selection using carbon footprint monitoring. This approach cuts across a huge variety of supplier base and caters to almost all businesses as environmentally friendly and robust. The approach encourages suppliers to go with green and to cut down their carbon footprints in order to survive the competition.

Hong-jun and Bin (2010) applied factor analysis identifying main factors for green supplier selection. The factor analysis is used to confirm the indices weights of each level, avoiding managerial subjectivity associated with tools such as AHP. Limitations include varying indices across different industries and sample bias which may affect the conclusions.

Feyzioglu and Buyukozkan (2010) proposed a multi-criteria evaluation model, using the Choquet Integral approach for supplier performance evaluation. The important feature of this operator is its ability to take into account criteria dependencies. The proposed method is more appropriate when compared to other methods that used only arithmetic means.

Awasthi et al. (2010) proposed an approach using three steps: identification of criteria, experts rating, and assessment of experts rating through a fuzzy TOPSIS methodology. The methodology ranks the supplier alternatives based on environmental performance, recommending selection of the highest valued alternative. An advantage of the fuzzy TOPSIS approach is its integration of benefit and cost criteria. Sensitivity analysis was conducted in this paper to evaluate the influence of criteria weights on the environmental performance evaluation of the suppliers. The approach is relatively practical and can provide a solution under partial lack of quantitative information.

Large and Thomsen (2011) developed a Structural Equation Model by degree of green criteria which explains the environmental performance and the purchasing performance. The model

was analyzed with SmartPLS 2.0 software using data collected by the survey.

Chiou et al. (2011) developed a structural equation model for the data analysis which is used to implement the proposal for green innovation in order to improve the environmental performance and to enhance the competitive advantage in the global market of the company.

4.2. Multi-criteria decision-making – integrated methodology approach

From the review, it has been found that only eight papers (22.2%) used integrated approaches (summarized in Appendix 1).

Humphreys et al. (2003a, b) implemented a Knowledge Based System using Case Based Reasoning. This approach integrates the environmental criteria into the green supplier selection process and provides the guidelines for purchasing managers to select suppliers from an environmental point of view.

Li and Zhao (2009) used a threshold method and grey correlation analysis for the assessment of an index system for the supplier of vehicle components. They use AHP to determine weight of factors for the comprehensive assessment. The grey relational analysis evaluates the correlation between the factors.

Yan (2009) proposed an integrated approach adopting a genetic algorithm and AHP to achieve green supplier optimization evaluation and selection. The author introduces a genetic algorithm combined with current weights of a supplier to dynamically adjust weights determined from AHP. The supplier evaluation indicators become more evident and optimized, while the system can dynamically adjust over time for a better supplier evaluation indicator.

Kuo et al. (2010) developed a green supplier selection model which integrates Artificial Neural Network (ANN), Multi Attribute Decision Analysis (MADA), Data Envelopment Analysis (DEA) and ANP. From the study, it has been discovered that ANN-MADA has better power of discrimination and noise insensitivity in evaluating a green supplier's performance. It also overcomes the DEA's drawbacks and limitations of data accuracy and decision making units among constraints.

Wen and Chi (2010) introduced DEA into the assessment of green supplier selection combined with AHP to establish an integrated model. This model overcomes the limitation of an individual approach of the traditional tools such as AHP and ANP for evaluation of suppliers. In this model, DEA filters suppliers to lessen the supplier set. AHP/ANP can then proceed more efficiently on the smaller subset.

Thongchattu and Siripokapirom (2010) applied AHP for green supplier selection, which allows the decision maker to structure complex problems. Added to the AHP model, the author proposed the consensus final decision using ANN.

Chen et al. (2010) apply fuzzy set theory accompanied with grey relational analysis for green supplier selection. The proposed method uses linguistic preference structures for deriving priorities for alternatives and uses grey numbers in all criteria and alternatives to avoid criteria limitations.

Kuo and Lin (2011) proposed a method which integrates ANP and DEA for green supplier evaluation. DEA and ANP consider the interdependency between the criteria. It expands on DEA by allowing users to restrict the weights using their own criteria weight preferences. This approach also allows for more flexibility on the number of decision making units (DMUs) used.

4.3. Environmental criteria based supplier selection

A summary of the environmental criteria used by the research identified in this study appear in Appendices 2–6. The most widely considered criterion for green supplier selection is “environmental

management system.” This major criterion is followed by green image, environmental performance, environmental competencies, design for environment, green competencies, corporate and social responsibilities, environmental efficiency, environmental authentication, environmental improvement cost, green logistic dimension, green organization activities, environmental certification, suppliers’ green image, use of environmentally friendly material, use of environmentally friendly technology, waste management, re-use, re-cycle, green process innovation, green product, green purchasing, green project partnership and green design. A very large set of criteria, some of which overlap, are utilized in various works.

Eleven papers (30.55%) used environmental management system as a criterion for supplier selection process. Environmental management system is typically considered a main criterion, with sub criteria such as environmental policies, environmental planning, and ISO14001 certification. These criteria are termed as qualitative environmental criteria and they required subjective decisions to be made during their evaluation. To summarize, most of the literature insists that the implementation of environmental management systems is a primary factor in identifying environmentally sound suppliers.

5. Observations and recommendations

In this review, thirty three journal articles related to green supplier evaluation and selection were collected and analyzed using multi-criteria decision-making approaches and evaluation criteria characteristics. We now present some observations and a somewhat critical analysis of these initial findings.

5.1. Widely used approaches

Researchers have developed and applied a variety of approaches for multi-criteria decision-making for the green supplier selection problem. One interesting characteristic is the preponderance of fuzzy analysis. Managerial and contextual uncertainty make for fuzziness in the decision making process, and ambiguity and intangibility cause difficulty in the analysis. We seek out the most popular approach adopted in the supplier evaluation and selection literature to provide insight into what is considered state-of-the-art. Interestingly, many of the identified papers – twenty five papers (77.77%) – are still utilizing a single technique in their analysis. This may be due to the ease of focusing on only one approach and by deliberately limiting the complexity of the approaches. Eight papers (22.22%) utilized an integrated approach, with the objective of trying to achieve a more realistic application given the complexities of a real-world decision process.

The most popular individual approach was AHP followed by ANP, DEA, and other tools. As the supplier selection problem involves both qualitative and quantitative criteria, AHP not only can handle adequately the inherent uncertainty and impression of human decision making process, but also it can provide the robustness and flexibility needed for the decision maker to understand the decision problem. AHP shows the performance of a supplier with respect to each sub criteria and main criteria, so it shows the supplier status on each criterion. Also, mathematically and philosophically, AHP provides an easily understandable and defensible approach to practitioners. It allows practitioners to be involved in the analysis and actually to guide the decision more effectively. This managerial transparency and lack of complexity allow for greater acceptance by both researchers and practitioners.

AHP is not without its critics, however, so ample consideration should be given to its limitations. The growth and application of AHP may derive more from a convenience and simplification perspective rather than from a strong theoretical mathematical perspective. The utilization and integration of fuzzy techniques may

be advantageous for some settings, but criticism has also been targeted to the overreliance on fuzzy mathematics for these decision environments. For example, some studies have shown that the ultimate decisions from fuzzified AHP models do not provide any differences in the ultimate solution than regular AHP. The additional complexity of utilizing fuzzy numbers may be unnecessary. Additional research on the ultimate solution quality under fuzzy and regular numerical valuations is necessary to determine whether the addition of fuzzy logic is a worthwhile effort. We conjecture that practitioners will choose basic AHP approaches over fuzzified approaches since regular AHP approaches are easier for the practitioners to understand and provide greater transparency. Investigating whether these additional complexities in modeling AHP are acceptable to practitioners is another avenue of potential research. Multi-Criteria Decision-Making approaches that were applied individually are summarized in [Appendix 7](#).

We also observe various integrated approaches for supplier selection. From the analysis, it was noticed that integrated AHP is still a very popular approach. The integration of AHP with other techniques may also be tied to its easy to understand mathematical basis, ease of use, and flexibility. AHP has been integrated with DEA, ANN, GA, and fuzzy set theory. The major reason for integration is that the individual techniques possess some unique advantages that allow for complementary contribution to the AHP approach. The consistency verification operation of AHP contributes greatly to prevent inconsistency because it acts as a feedback mechanism for the makers to review and revise the judgment. The Multi-Criteria Decision-Making approach which was applied in an integrated way is summarized in [Appendix 8](#).

Yet, multiple techniques, especially for the unversed, can cause lack of practitioner acceptance due to their increased complexity. But some integrative approaches may overcome these limitations by actually simplifying the ‘upfront’ process. For example, an initial filtration process reduces the number of alternatives and criteria, and may allow for lessened practitioner involvement. This type of situation is most evident when practitioner input is not needed for initial or later stages of multiple method techniques. Additional investigation into how multi-method techniques can be barriers or enablers to acceptance of certain integrative tools is certainly an important direction for future research.

5.2. Widely used evaluation criterion

In most of the recent studies, both traditional and ecological criteria are considered for supplier selection. The second objective of this paper is to discover the most popular criteria considered by the literature for evaluating and selecting the appropriate supplier. Hundreds of criteria were proposed and they are summarized in [Appendix 9](#).

Quantitative environmental criteria articulated in economic terms and qualitative environmental criterion which focuses on more intangible (e.g. company image and reputation) have been used for supplier selection ([Humphreys et al., 2003a, b](#)). We have observed in our data set that the major criteria considered are environmental management system and quality ([Grisi et al., 2010](#)). Some studies are also based on potential drivers like GSCM capabilities, strategic level of purchasing department, the level of environmental commitment, degree of green supplier assessment and degree of green supplier collaboration ([Large and Thomsen, 2011](#)).

In portfolio based analyses, pollution control and prevention are considered as environmental factors ([Lee et al., 2009](#)). There is a study in effect with four drivers namely regulation, customer pressure, social responsibility, and business benefits on green purchasing ([Buyukozkan and Cifci, 2010, 2011](#)). [Chen et al. \(2010\)](#) states that companies must engross suppliers and purchasers to

congregate and even exceed the environmental expectation of their customers and their governments. Many manufacturing concerns organize through the outsourcing and thus the companies are vitally dependent on their suppliers. Most of the companies ask their suppliers to implement ISO14001 since the standard becomes a prevalent tool for environmental aspects and factors in the sustainable supply chain (social, economic and environmental) (Chen et al., 2010; Buyukozkan and Cifci, 2011).

The prevalence and shifting focus of environmental criteria add additional complexities for both practitioners and researchers. Although we have mentioned some studies that try to target certain criteria, especially environmental ones, consensus criteria are difficult to come by. Jabbour and Jabbour (2009) analyzed the inclusion of environmental criteria in the supplier selection process at organizations; they verified that the organizations have difficulties including environmental aspects in the supplier selection process. Clearly, more generic and encompassing criteria, e.g. environmental management systems implementation, are more acceptable due to their flexibility. The difficulty in these situations is that their generality also leads to greater ambiguity in what criteria actually mean. Care needs to be taken when introducing these intangible and broad criteria, which cause greater uncertainty and trust in the data and results. It can also be frustrating to those suppliers who are influenced by the decisions these models recommend.

Additional research is required to identify and to more clearly define each of the criteria. Building accepted definitions and characteristics of these criteria before their implementation in decision models is necessary for the acceptance of the methodologies, and, most importantly, for the decisions that derive from these methodologies. For example, the utilization of exploratory factor analysis of practitioners may be a way to help determine and categorize factors into various groupings. Also, methods for identifying, defining, grouping, and filtering criteria would greatly enhance the research, development, and application of models and their criteria. These areas are important and necessary directions.

6. Results and future work

From the review, it has been found that individual methodological approaches were used more than multiple integrated methodological approaches and environmental criterion were not precisely focused in many articles. Also, the weightings of supplier evaluating criteria depend on business priorities and strategies. In cases where the weightings are assigned arbitrarily and subjectively, this leads to the supplier selection that may not be accurate based on company requirements. Many approaches work for the primary selection of a suitable supplier, but do not necessarily recommend why that supplier is the best choice or why suppliers who fail in the selection process might improve their performance standings.

We also observed that sensitivity analysis was not completed in many circumstances. Sensitivity analysis investigates the impacts of criteria weights on the selection of supplier with best environmental performance by changing the weights of the criteria for several experiments. So, it has to be carried in future multi-criteria decision-making approaches. As part of a sensitivity analysis, the careful consideration of various stakeholder and multi-functional influences need integration. The unique characteristics of environmental and general sustainability dimensions means that inputs from a variety of stakeholders (e.g. communities, supplier's suppliers, environmental consultants, and staff) may be necessary to help address some of the more sensitive issues facing environmental performance and criteria of suppliers.

Also, we have identified in the previous section additional avenues for further investigation, focusing primarily on the methodologies and criteria. These investigations need to include the level of

acceptance of models by practitioners and researchers. Comparative analyses truly need to be completed. This comparative analysis needs to be both research and practically oriented. Validation, acceptance, and reliability of techniques are required for additional development. These investigations can clarify strengths and weaknesses that may not be observable to the developers and modelers.

Many researchers, even ourselves, may be biased for one approach or another. For scientific progress, additional investigations in experimental settings are needed, especially as the complexities of the environmental and ecological factors come into play. Interesting organizational and behavioral decision studies need to follow up on these prescriptive approaches. For example, longitudinally are these results associated with the greenest suppliers that are selected, confirmed months or years after the selection? Are the decisions that are made good ones? This type of research requires significant resources, patience, and a considerable degree of follow-up. It may also require unprecedented access to the decision maker and companies. But, if, as researchers, we are to strengthen our modeling efforts, we need to carefully examine these directions.

7. Conclusion

This paper reviewed multi-criteria decision making approaches for supplier evaluation and selection in literature from 1997 to 2011. Many individual and integrated approaches have been proposed for supplier selection. The literature review has highlighted that the implementation of green issues within the supplier selection process is limited as relatively few papers were identified. We found that the most widely used multi-criteria decision-making approach is analytical hierarchy process (AHP) and the most widely considered criteria for green supplier evaluation and selection is environmental management system.

Besides some recommendations like crisp results in supplier selection, failure mode analysis for failed suppliers might be added to aid the researchers and decision makers in solving the green supplier selection problem effectively. Additionally, a sensitivity analysis should be conducted as a primary method of selecting a supplier with the best environmental performance; it should take into account changing weights and it should address sensitive data changes so that choosing the wrong supplier can be avoided. We have also identified a number of potential, open ended, developmental, and research questions that focus on the validity of the models and criterion. The location application has to be tested and carried out to find the fitness of research. Substantial research in the behavioral and applied characteristics of these models is needed. This study contains some limitations, such as the extensiveness of the reviewed literature and the inclusion of the proceedings articles which were not peer reviewed.

The critical analysis we provide needs to tie the model developers with behavioral decision making literature. Experimental designs, not just application validation, are needed for future research. We could have tied in these additional multiple criteria decision analysis concerns, which may be even more evident in socially sensitive decisions such as green supplier selection.

There are ample opportunities for future investigations in a number of directions for this research. We hope established and younger scholars investigate many of these issues as the problems we face from an environmental perspective infringe on our lives more than ever.

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Appendix 1. Summary of literature on multiple criteria modeling and green supplier selection and evaluation.

S. no	Authors	Approach	Author view about this approach	Relationship to this study
1	Noci (1997)	AHP	This approach seeks to identify a critical supplier's environmental performance within pro-active 'green' strategies to allow the decision-maker to develop a complete vendor rating procedure.	This article projects the supplier's environmental performance through vendor rating.
2	Handfield et al. (2002)	AHP	To evaluate the relative importance of various environmental traits and to access the relative performance of several suppliers along these traits.	This paper applies the environmental criteria to supplier assessment.
3	Chiou et al. (2008)	FAHP	To solve the green supplier selection problem by a ranking system based on different weights of various criteria.	Involves four environmental criteria among six major criteria.
4	Lu et al. (2007)	FAHP	To capture the expertise of experienced supply chain designers, easily computerized and to handle a variety of environmental situation effectively.	This paper applies the environmental principles to green supplier selection.
5	Grisi et al. (2010)	FEAHP	This approach allows better management of data involved in global decisions, covers the use and integration of quantitative and qualitative data, provides the necessary flexibility for the analysis of problem, and facilitates tasks of verification for the robustness of the decision taken.	Supplier performance and evaluation is based in green supply chain management.
6	Lee et al. (2009)	FEAHP	To consider the vagueness of experts opinion and importance weightage of criteria, the fuzzy extended analytic hierarchy process exploited.	Environmental criteria were mainly considered for supplier selection.
7	Buyukozkan and Cifci (2010, 2011)	FANP	This method not only makes sufficient evaluation using the provided preference information, but also maintains the consistency level of evaluation	Uses social responsibility and environmental competencies as factors for sustainable supplier selection.
8	Hsu and Hu (2007, 2009)	ANP	This method is characterized by interdependencies among decision structure components.	It involves application of hazardous substance management to supplier selection.
9	Buyukozkan and Cifci (2010, 2011)	FANP	The dynamic characteristics and complexity of GSCM analysis environment makes the ANP technique as a suitable tool. To cope with ambiguity and vagueness of the decision maker's evaluation, the fuzzy extension of the ANP method is preferred.	This study has a goal to improve the green supply chain management activities.
10	Chiou et al. (2011)	Confirmatory Factor Analysis	The data collected through survey based on hypothesis was analyzed using confirmatory factor analysis; final measurement models are used to evaluate the structural model that verifies the significance of proposed relationship.	This paper influences the supplier towards green issues and innovation on environmental performance.
11	Yeh and Chuang (2011)	MOGA	In order to solve the conflicting objectives, the multi objective genetic algorithm is used to find the Pareto-optimal solutions, which utilized the weighted sum approach that can generate a greater number of solutions.	Supplier selection purely based on green criteria.
12	Large and Thomsen (2011)	Structural Equation Model	It can be analyzed by SmartPLS 2.0 software using data collected through survey.	Mainly considers environmental criteria for measurement assessment.
13	Vachon and Klassen (2006)	Chi Square Test	In this study, the data from survey was used for testing the linkage between green project partnership and five performance indicators.	It involves the green project partnership with suppliers.
14	Feyzioglu and Buyukozkan (2010)	Choquet integral	The important feature of this operator is its ability to take into account the criteria dependencies. The proposed method is showed to be more appropriate compared to other methods based on arithmetic mean.	Considers the environmental performance criteria for supplier selection model.
15	Kumar and Jain (2010)	DEA	This model does not demand exact weights from decision maker. The comparative values of decision making units assess the supplier evaluation.	Green approaches with carbon footprint monitoring.
16	Yang and Wu (2008)	Extensible Synthetic Model	This model overcomes the lower weight factor avoidance and the result is more objective than other evaluation methods.	Mainly focuses on green suppliers.
17	Hong-jun and Bin (2010)	Factor Analysis	The factor analysis is used to confirm the indices weights of each level where the method can avoid the subjectivity of AHP.	Selection of supplier based on green purchasing.

(continued on next page)

Appendix 1 (continued)

S. no	Authors	Approach	Author view about this approach	Relationship to this study
18	Humphreys et al. (2006)	Fuzzy Inference Method	The major benefits of this system are that in a computationally inexpensive manner the proposed system is capable of implementing a range of user priorities that influence to varying degrees the system output.	Assess environmental performance in the supplier selection process.
19	Awasthi et al. (2010)	Fuzzy Topsis	The advantages of using fuzzy TOPSIS is that it distinguishes between benefit and cost category criteria.	Evaluation of supplier based on environmental performance and fuzzy multi-criteria based approach.
20	Yang and Wu (2008)	Grey Entropy Synthetic Evaluation Model	This model overcomes the lower weight factor avoidance and the result is more objective than other evaluation methods.	Supplier selection based on environmental criteria.
21	Humphreys et al., 2003a, b	Knowledge Based System	The decision support system which integrates the environmental criteria into the supplier selection process was built and provides guidelines for the purchasing managers to select suppliers from an environmental viewpoint.	Evaluating the supplier environmental performance.
22	Bai and Sarkis (2010a)	Rough Set Theory	Rough set methodology which utilizes an incomplete information approach which is more realistic in source data poor environments for green supplier development.	Focus on green supplier development.
23	Zhang et al., (2003)	Fuzzy Multi Agent Decision Making Model	It is used to evaluate the life cycle environmental impact of purchased product.	Focus in environmentally conscious supplier management.
24	Bala et al. (2008)	Cross Case Analysis	A comparison of the different cases is carried out and some general rules are created for evaluation.	Case study – Experience with green supplier.
25	Thongchattu and Siripokapirom (2010)	AHP and ANN	Analytical Hierarchy Process (AHP) for green supply chain supplier selection process allows the decision maker to structure complex problems. Added to AHP model, the final decision by Artificial Neural Network is improved by minimizing a limitation error set.	The framework involves the environmental project and standard for environmental management system.
26	Yan (2009)	AHP and GA	The common way of achieved green supplier selection was to accomplish through matrix with AHP. But weights are static and there is no dynamic optimization adjustments mechanism, so the author introduces genetic algorithm combined with current weights of supplier to dynamically adjust weights.	It is research on green supplier evaluation.
27	Kuo et al. (2010)	ANN-MADA	ANN-MADA has better power of discrimination and noise insensitivity in evaluating green supplier's performance. It also overcomes the DEA drawbacks and limitation of data accuracy and decision making units among constraints.	Supplier selection based on environmental criteria.
28	Kuo and Lin (2011)	DEA and ANP	As per authors' opinion, this model of DEA and ANP considers the interdependency between criteria. This also removes the constraints of DEA that users cannot set up their own criteria weight preferences. The proposed evaluation method also solves the limitation of the number of decision making units (DMUs).	Supplier selection based on environmental criteria.
29	Wen and Chi (2010)	DEA with AHP and ANP	The advantage of this model was to overcome the limitation of individual approach of traditional tools such as AHP and ANP for evaluation of supplier. In this model DEA distinguishes the efficient candidates of supplier from the entire group. So AHP/ANP can proceed with further analysis without considering unnecessary suppliers.	Developing green supplier selection procedure.
30	Chen et al. (2010)	Fuzzy Set Theory and Grey Relational Analysis	The fuzzy set theory is a mathematical theory designed to model the fuzziness of human cognitive process. This study involves uncertainty, the proposed method uses the linguistic preferences for deriving the priorities of different selection alternatives, and sets up grey numbers in all criteria and alternatives to avoid the negligence of same criteria.	It implements the green supply chain management.
31	Li and Zhao (2009)	Grey Correlational Analysis and AHP	Authors adopt AHP to determine weight of factors as they are of fundamental importance for comprehensive assessment. The grey relational analysis measures the size of correlation between the two factors.	Considers environmental factors for the supplier selection.
32	Humphreys et al., 2003a, b	Knowledge based system and Case based reasoning	Knowledge Based System with subset of Case Based Reasoning which integrates the environmental criteria in the green supplier selection process and provides the guidelines for purchasing managers to select suppliers from an environmental viewpoint.	It evaluates the environmental management performance of the supplier.

Appendix 2. Summary of AHP application and criteria used for green supplier selection and evaluation.

Analytical hierarchy process - application and evaluating criteria								
S. No	Authors	Application	Evaluating criteria					
			Main criteria	Sub criteria				
1	Noci (1997)	Automobile Industry	Green Competencies	Availability of green technologies	Type of Material used in the supplied component	Capacity to respond in time		
			Current Environmental efficiency	Air emission	Solid Wastes	Waste Water	Energy consumption	
			Supplier's Green image	Customers purchase retention	Type of relationship with stakeholders	Market share related to green customers		
			Net Life cycle cost	Cost of the supplied component	Cost of component disposal	Depreciation for improvement		
2	Handfield et al. (2002)	Automotive, Apparel and Paper Manufacturer	Product attributes					
			Waste management					
			Labeling/certification					
			Packaging/reverse logistics					
			Compliance to Government Regs.					
			Environmental Pgms at the supplier's facilities					
3	Chiou et al. (2008)	Electronic Industry	Supply chain base	Cost	Quality	Delivery schedule	Service performance	
			Green competencies	Green material selection	Cleaner production technologies	Reduced green packaging	Reverse logistics	
			Environmental management systems	ISO-14001 certification	Eco-labeling	Supplier environmental evaluation	Environmental management information system	
			Environmental performance	Green design	Use of toxic/restricted substances	Remanufacturing/reuse activities	Waste management	
			Corporate social responsibilities	Public disclosure of environmental record	Green image	Relationship with stakeholders	Contributions to community	
			Risk factor	Government & local rules/regulations	Political stability	Suppliers' credit	Internal management system	Gaseous residue
4	Lu et al. (2007)	Electronics Industry	Environmental criteria	Materials	Energy use	Solid residue	Liquid residue	End of life
			Supply chain stages	Premanufacturing	Product manufacture	Distribution, packaging	Product use, maintenance	
5	Grisi et al. (2010)		Price	Price of goods	Shipping costs	Costs of pollution effects	Rates and customs taxes	
			Quality of Delivery	Response to specific requests of the company	Timeliness			
			Quality	Increased rate of return of the product	Increased average waiting time	Processes for internal audit quality		
			Environmental Competences	Availability of "clean" technologies	ecological materials	Ability to respond to reduce the environmental impact		
			Environmental Management System	Environmental policies	Environmental Planning	Implementation and operation	ISO14001	
			Green Image	Segment of green customers	Type of relationship	Customers fidelization		
			Current Environmental Impact	Emission of air pollution	Substances discharged into the sewer	Solid waste produced	Power consumption	

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Appendix 2 (continued)

Analytical hierarchy process - application and evaluating criteria							
S. No	Authors	Application	Evaluating criteria				
			Main criteria	Sub criteria			
6	Lee et al. (2009)	Electronics industry	Quality	Quality-related certificates	Capability of quality management	Capability of handling abnormal quality	
			Technology capability	Technology level	Capability of R&D	Capability of design	Capability of preventing pollution
			Total product life cycle cost	Cost of component disposal			
			Green image	Ratio of green customers to total customers	Social responsibility		Use of harmful materials
			Pollution control	Air emissions	Waste water	Solid wastes	Energy consumption
			Environment management	Environment-related certificates	Continuous monitoring and regulatory compliance	Green process planning	Internal control process
			Green product competencies	Recycle	Green packaging		
				Materials used in the supplied components that reduce the impact on natural resources	Ability to alter process and product for reducing the impact on natural resources		

Appendix 3. Summary of analytical network process – application and evaluating criteria for green supplier selection and evaluation.

S. No	Authors	Application	Evaluating criteria					
			Main criteria	Sub criteria				
1	Buyukozkan and Cifci (2010, 2011)	Commercial Industry-White Goods Company in Turkey	Organization Financial Performance Service Quality Technology Social responsibility & Environmental competencies					
2	Hsu and Hu (2007)	Electronic Industry	Procurement Management	Requirement of green purchasing	Green materials coding and recording	Inventory of substitute material	Supplier Management	
			R&D Management	Capability of green design	Inventory of hazardous substances	Legal-compliance competency		
			Process Management	Management for hazardous substances	Prevention of mixed material	Process auditing	Pre-shipment inspection	Warehouse management
			Incoming Quality Control	Standard for incoming quality control	Test equipment	Record of incoming quality control		
			Management System	Quality control management for system	Environmental management for system	Hazardous substance management for system	Information Systems	

Appendix 3 (continued)

S. No	Authors	Application	Evaluating criteria						
			Main criteria		Sub criteria				
3	Hsu and Hu (2009)	Electronic Industry	Procurement management	Requirement of green purchasing	Green materials coding and recording	Inventory of substitute material	Suppliers management		
			R&D management	Capability of green design	Inventory of hazardous substances	Legal-compliance competency			
			Process management	Management for hazardous substances	Prevention of mixed material	Process auditing	Pre-shipment inspection	Warehouse management	
			Incoming quality control	Standard for incoming quality control	Test equipment	Record of incoming quality control			
			Management system	Quality management system	Environmental management system	Hazardous substance management system	Information Systems		
4	Buyukozkan and Cifci (2010, 2011)	Manufacturing Industry	Organizational performance dimension	Quality	Cost	Time	Flexibility		
			Green Logistics Dimension	Procurement	Production	Distribution	Reverse L.	Packaging	
			Green organizational activities dimension	Reduce	Recycle	Remanufacture	Reuse	Disposal	
			Alternative green supply chain systems, projects, practices, etc.	Current system	System A	System B			

Appendix 4. Mathematical programming approaches – application and evaluating criteria for the green supplier evaluation and selection process.

S. No	Approaches	Authors	Application	Evaluating criteria
1	MOGA	Yeh and Chuang (2011)	Electronics Industry	Cost Transportation Time Yield Air pollution treatment cost

Appendix 5. Other approaches – application and evaluating criteria.

S.No	Approaches	Authors	Application	Evaluating criteria					
				Main criteria	Sub criteria				
1	Chi Square test	Vachon and Klassen (2006)	Package Printing Industry	Manufacturing performance	Cost	Quality	Delivery	Flexibility	Environmental
				Green projects partnership	Partnership with suppliers	Partnership with customers			
2	Choquet integral	Feyzioglu and Buyukozkan (2010)	White good producer	Contextual variables	Plant size	Parent company size	Reinvestment rate	Age of presses	Supplier base
				Product Cost	Product price	Freight cost	Tariff and custom duties		
				Product Quality	Rejection rate of the product	Increased lead time	Quality assessment	Remedy for quality problems	
				Service Performance	Delivery performance	Technology and RD support	Responsiveness	Ease of communication	Supplier reputation
				Environmental Performance	Environmental management competencies	Existing environmental management systems	Design for environment	Production for environment	Logistics for environment
3	DEA	Kumar and Jain (2010)		Carbon footprint	Environmental costs				

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Appendix 5 (continued)

S.No	Approaches	Authors	Application	Evaluating criteria					
				Main criteria	Sub criteria				
4	Extensible synthetic model	Yang and Wu (2008)	Electrical appliances manufacturing	Quality	Quality system	Process capability exponent	Total quality management	Rate of certified product	
				Price					
				Capability of suppliers	Supplying capability	Level of technique	Capability of product development	Influence degree in market	P/E ratio
				Service	Accurate rate of processing order	Rate of delivery	Time to solve complaint	Degree of information modernized	Service manner
				Environment protection	ISO14001 certification	Green design of product	Validity of clean technique	Environment efficiency	
				Management Systems and Credit Standing	Enterprise systems	Management systems	Management ideas	Researching and developing direction	Enterprise' credit standing
5	Factor analysis	Hong-jun and Bin (2010)	Manufacturing Industry	Supply chain management	Supply in time	Strategic cooperation	Quality	Credit	Financial status of supplier
				Supply chain management	Solving problems with suppliers	Cost of product	Complaint for product	Training for employee	Culture compatibility between company
				Supply chain management	Communication with other companies				
				Eco-design	Reuse of package when design	Reduction the use of hazard materials when design	Rebuild of products when design	Reduction of package when design	Recycle, reuse and resume of products when design
				Operation management	Innovation of technique	Productivity	ISO9000 certification		
				Outside environmental management	Waste of water	Exhaust gas	Recycle system for wastes	Dealing with hazard solid waste	Environmental friendly behavior assessment to second-tier supplier
6	Fuzzy inference method	Humphreys et al. (2006)	Toxic Gas Database	Environmental authentication	ISO14001 certification	Eco-label of products	Air Emission	Waste water disposal	Energy
				Environmental-Pollutant Effects	Solid Waste	Chemical Waste			
				Environmental-Improvement Costs	Buying new friendly material	Buying new environmentally friendly technology	Redesign of product	Staff Training	Recycling
				Green Image	Customer Purchasing retension	Green Market Share	Stakeholders Relationship		
				Design for Environment	Recycle	Reuse	Re Manufacture	Disassembly	Disposal
				Env.Mgmt System	Env.Policies	Env.Planning	Implement and operation	ISO14001 Certification	
				Environmental Competencies	Clean Technology availability	Use of Env. Friendly material	Pollution reduction capability	Returns handling capacity	
7	Fuzzy topsis with sensitivity analysis	Awasthi et al. (2010)	Logistics	Use environment friendly technology					
				Us of environmental friendly material					
				Green market share					
				Partnership with green organization					
				Management commitment					
				Adherence to environmental policies					
				Green R & D Projects					
				Staff Training					
				Lean Process Planning					

Appendix 5 (continued)

S.No	Approaches	Authors	Application	Evaluating criteria					
				Main criteria	Sub criteria				
8	Grey entropy synthetic evaluation model	Yang and Wu (2008)	Electrical appliances manufacturing	Design for Environment					
				Environmental certification					
				Pollution control initiatives					
				Quality	Quality system	Process capability exponent	Total quality management	Rate of certified product	
				Price					
				Capability of suppliers	Supplying capability	Level of technique	Capability of product development	Influence degree in market	P/E ratio
9	Knowledge based system	Humphreys et al., 2003a, b		Service	Accurate rate of processing order	Rate of delivery	Time to solve complaint	Degree of information modernized	Service manner
				Environment protection	ISO14001 certification	Green design of product	Validity of clean technique	Environment efficiency	
				Management Systems and Credit Standing	Enterprise systems	Management systems	Management ideas	Researching and developing direction	Enterprise' credit standing
				Environmental-Pollutant Effects	Solid Waste	Chemical Waste	Air Emission	Waste water disposal	Energy
				Environmental-Improvement Costs	Buying new friendly material	Buying new environmentally friendly technology	Redesign of product	Staff Training	Recycling
				Management Competencies	Senior management support	Environmental Partners	Training	Information Exchange	
10	Rough Set Theory	Bai and Sarkis (2010)	Hypothetical application	Green Image	Customer Purchasing retension	Green Market Share	Stakeholders Relationship		
				Design for Environment	Recycle	Reuse	Re Manufacture	Disassembly	Disposal
				Env.Mgmt System	Env. Policies	Env. Planning	Implement and operation	ISO14001 Certification	
				Environmental Competencies	Clean Technology availability	Use of Env. Friendly material	Pollution reduction capability	Returns handling capacity	
				Green Knowledge Transfer					
				Investment and Resource transfer					
11	Fuzzy Multi agent decision making model	Zhang et al., (2003)	University Application	Management and organizational practices					
				Profitability attributes					
				Cost					
				Flexibility					
				Quality					
				Environment					
12	Cross case analysis	Bala et al. (2008)	University Application	Recycle					
				Reuse					
				Service					
				Strategic level of purchasing					
				Environmental commitment					
				Purchasing's environmental capabilities					
13	Structural equation model	Large and Thomsen (2011)		Green supplier assessment					
				Green collaboration with suppliers					
				Environmental performance improvement					
				Purchasing performance					
				Greening the supplier					
				Green product innovation					
14	Confirmatory factor analysis	Chiou et al. (2011)	Survey on Many Industry	Green process innovation					
				Green managerial innovation					
				Environmental performance					
				Competitive advantage					

Appendix 6. Integrated approaches – application and evaluating criteria

S.No	Approaches	Authors	Application	Evaluating criteria					
				Main criteria	Sub criteria				
1	AHP and ANN	Thangchattu and Siripokapiram (2010)		Company reliability	Capacity and lead time				
				Material quality	Defect				
				Material price					
				Environmental	Green project				
				ISO 14000	Environmental Performance Evaluation	Environmental labeling	Auditing	Life Cycle Assessment	ISO 14000 Certificate
2	AHP and GA	Yan (2009)		Environment	The usage of raw materials	The waste recovery disposal	Pollution control		
				Quality	Product competitiveness				
				Operational capacity	Rapid response capability	Delivery punctual rate	Delivery cycle		
				Price level	Transaction costs	Price level			
				Service level	Service attitude	After service	Information sharing		
3	ANN-MADA	Kuo et al. (2010)	Digital Camera Manufacturing Company in Taiwan	Quality	Reject rate	Management commitment of quality	Process improvement	Warranties and claim policies	Quality assurance
				Cost	Price performance value	Compliance with sectoral price behavior	Transportation cost		
				Delivery	Order fulfillment rate	Lead Time	Order frequency		
				Service	Responsiveness	Stock management	Willingness		
				Environment (Env.)	EUP (Eco-design requirements)	ODC- ozone depleting chemicals	RoHS - Restriction of hazardous substance	Design capability ISO-14001 - certification requirement	WEEE- waste electrical electronic equipment
				Corporate Social Responsibility (CSR)	The interest and right of employee	The rights of stakeholder	Information disclosure	Respect for the policy	
4	DEA and ANP	Kuo and Lin (2011)	High-tech Industry	Organization structure and manufacturing capability	Technological capability	Manufacturing capability	Financial status		
				Supplier's implementation capability	Delivery quality	Delivery schedule	Price	Cooperation	
				Quality system	Supply quality assurance	Inspection situation of quality control	Implementation situation of quality control	Manufacturing process improvement	
				Environmental issues	Environmental administration system	Environmental system	Environmental planning	Green purchasing	
5	DEA with AHP and ANP	Wen and Chi (2010)		Green management system					
				Supplier's green image					
				Green product performance					
				Financial issues					
				Quality					
				Service					
				Technology issues					
				Organizational culture & strategy issues					

Appendix 6 (continued)

S.No	Approaches	Authors	Application	Evaluating criteria				
				Main criteria	Sub criteria			
6	Fuzzy Set Theory and Grey Relational Analysis	Chen et al. (2010)	Electronics Industry-PCB Firm	Delivery reliability Profitability of supplier Supplier relationship Conformance quality Flexibility of supplier Internal service quality Green design Green purchasing Life cycle assessment ISO 14000 certificates Internal green production plan Management support Cleaner production Environmental management system R&D capability R&D green products Quality				
7	Grey Correlational Analysis and AHP	Li and Zhao (2009)	Electronics Company	Technology Service Innovation Management Environment	Quality Assurance System Product Performance Time Delivery Rate Proportion of Engineers and Technicians Compatibility of Strategic objective Geographic Location of Supplier Solid Waste	Product Passing Rate Production Agility Level of Maintenance Service. Proportion of R&D Accounted for Sales Revenue Compatibility of Organization Relationship between Supplier and Community Chemical Waste	Green Degree in the Process of Product Manufacturing, Use, and Recycling Green Degree of Manufacturing Process Service Attitude. Green Degree in Innovation Behavior. Compatibility of Management Level Position and Reputation of Supplier in the Air emission	Extent of Information Standardization Security and Compensation Compatibility of Corporate Culture
8	Knowledge based system and Case based reasoning	Humphreys et al., 2003a, b	Telecommunication company	Environmental-Pollutant Effects Environmental-Improvement Costs Management Competencies Green Image Design for Environment Env.Mgmt system Environmental competencies	Buying new friendly material Senior management support Customer Purchasing retention Recycle Env. Policies Clean Technology availability	Buying new environmentally friendly technology Environmental Partners Green Market Share Reuse Env. Planning Use of Env. Friendly material	Redesign of product Training Stakeholders Relationship Re Manufacture Implement and operation Pollution reduction capability	Waste water disposal Staff Training Information Exchange Disassembly ISO14001 Certification Returns handling capacity
								Energy Recycling Disposal

Appendix 7. Summary of individual approaches

Approaches	No. of articles	Authors
Analytical hierarchy process	6	Noci (1997); Handfield et al. (2002); Chiou et al. (2008); Lu et al. (2007); Grisi et al. (2010); Lee et al. (2009)
Analytical network process	4	Buyukozkan and Cifci (2010); Hsu and Hu (2007, 2009); Buyukozkan and Cifi (2011)
Chi square test	1	Vachon and Klassen (2006)
Choquet integral	1	Feyzioglu and Buyukozkan (2010)
Confirmatory factory analysis	1	Chiou et al. (2011)
DEA	1	Kumar and Jain (2010)
Extensible synthetic model	1	Yang and Wu (2008)
Factor analysis	1	Hong-jun and Bin (2010)
Fuzzy inference method	1	Humphreys et al. (2006)
Fuzzy topsis with sensitivity analysis	1	Awasthi et al. (2010)
Grey entropy synthetic evaluation model	1	Yang and Wu (2008)
Knowledge based system	1	Humphreys et al. (2006)
MOGA	1	Yeh and Chuang (2011)
Rough set theory	1	Bai and Sarkis (2010a; b)
Structural equation model	1	Large and Thomsen (2011)
Fuzzy multi agent decision making model	1	Zhang et al., (2003)
Cross case analysis	1	Bala et. al (2008)
Total	25	

Appendix 8. Summary of integrated approaches

Approaches	No. of articles	Authors
AHP and ANN	1	Thangchattu and Siripokapiram (2010)
AHP and GA	1	Yan (2009)
ANN-MADA	1	Kuo et al. (2010)
DEA and ANP	1	Kuo and Lin (2011)
DEA with AHP and ANP	1	Wen and Chi (2010)
Fuzzy Set Theory and Grey Relational Analysis	1	Chen et al. (2010)
Grey Correlational Analysis and AHP	1	Li and Zhao (2009)
Knowledge based system and Case based reasoning	1	Humphreys et al., 2003a, b
Total	8	

Appendix 9. Summary of evaluating criteria

Criteria	No. of articles	Author
Adherence to environmental policies	1	Awasthi et al. (2010)
Air pollution treatment cost	1	Yeh and Chuang (2011)
Alternative green supply chain systems, projects, practices, etc.	1	Buyukozkan and Cifci (2010, 2011)
Capability of suppliers	1	Yang and Wu (2008)
Carbon footprint	1	Kumar and Jain (2010)
Cleaner production	1	Chen et al. (2010)
Company reliability	1	Thangchattu and Siripokapiram (2010)

Appendix 9 (continued)

Criteria	No. of articles	Author
Competitive advantage	1	Chiou et al. (2011)
Compliance to Government regs.	1	Handfield et al. (2002)
Conformance quality	1	Chen et al. (2010)
Contextual variables	1	Vachon and Klassen (2006)
Corporate social responsibilities	2	Chiou et al. (2008); Kuo et al. (2010)
cost	3	Yeh and Chuang (2011); Kuo et al. (2010); Zhang et al., (2003)
Current Environmental efficiency	2	Noci (1997); Grisi et al. (2010)
Delivery	2	Kuo et al. (2010); Chen et al. (2010)
Design for Environment	3	Awasthi et al. (2010); Humphreys et al. (2006); Humphreys et al., 2003a, b; Hong-jun and Bin (2010)
Eco-design	1	Hong-jun and Bin (2010)
Env. Mgmt System	11	Humphreys et al. (2006); Humphreys et al., 2003a, b; Thangchattu and Siripokapiram (2010); Li and Zhao (2009); Yan (2009); Zhang et al., (2003); Kuo et al. (2010); Lee et al. (2009); Chen et al. (2010); Grisi et al. (2010); Chiou et al. (2008)
Environment protection	1	Yang and Wu (2008)
Environmental authentication	2	Hong-jun and Bin (2010); Awasthi et al. (2010)
Environmental commitment	1	Large and Thomsen (2011)
Environmental Competences	3	Grisi et al. (2010); Humphreys et al. (2006); Humphreys et al., 2003a, b
Environmental criteria	1	Lu et al. (2007)
Environmental issues	1	Kuo and Lin (2011)
Environmental performance	3	Chiou et al. (2011); Chiou et al. (2008); Feyzioglu and Buyukozkan (2010); Large and Thomsen (2011)
Environmental Pgms at the supplier's facilities	1	Handfield et al. (2002)
Environmental-Improvement Costs	2	Humphreys et al. (2006); Humphreys et al., 2003a, b
Financial issues	1	Wen and Chi (2010)
Financial Performance	2	Buyukozkan and Cifci (2010, 2011)
Flexibility	2	Zhang et al., (2003); Chen et al. (2010)
Green product performance	1	Wen and Chi (2010)
Green collaboration with suppliers	1	Large and Thomsen (2011)
Green competencies	3	Lee et al. (2009); Noci (1997); Chiou et al. (2008)
Green design	1	Chen et al. (2010)
Green image	4	Lee et al. (2009); Humphreys et al. (2006); Humphreys et al., 2003a, b; Grisi et al. (2010)
Green image	1	Humphreys et al. (2006)
Green knowledge transfer	1	Bai and Sarkis (2010a)
Green logistics dimension	2	Buyukozkan and Cifci (2010, 2011)
Green management system	1	Wen and Chi (2010)
Green managerial innovation	1	Chiou et al. (2011)
Green market share	1	Awasthi et al. (2010)
Green organizational activities dimension	2	Buyukozkan and Cifci (2010, 2011)
Green process innovation	1	Chiou et al. (2011)
Green product	1	Lee et al. (2009)

Appendix 9 (continued)

Criteria	No. of articles	Author
Green product innovation	1	Chiou et al. (2011)
Green projects partnership	1	Vachon and Klassen (2006)
Green purchasing	1	Chen et al. (2010)
Green R & D Projects	1	Awasthi et al. (2010)
Green supplier assessment	1	Large and Thomsen (2011)
Greening the supplier	1	Chiou et al. (2011)
Incoming quality control	1	Hsu and Hu (2007, 2009)
Innovation	1	Li and Zhao (2009)
Internal green production plan	1	Chen et al. (2010)
Internal service quality	1	Chen et al. (2010)
Investment and Resource transfer	1	Bai and Sarkis (2010a)
ISO 14000	2	Thangchattu and Siripokapiram (2010); Chen et al. (2010)
Labeling/certification	1	Handfield et al. (2002)
Lean process planning	1	Awasthi et al. (2010)
Life cycle assessment	1	Chen et al. (2010)
Management	1	Li and Zhao (2009)
Management and organizational practices	1	Bai and Sarkis (2010a,b)
Management commitment	1	Awasthi et al. (2010)
Management competencies	2	Humphreys et al. (2006); Humphreys et al., 2003a, b
Management support	1	Chen et al. (2010)
Management system	2	Hsu and Hu (2007, 2009)
Management System (D5)	2	Hsu and Hu (2007, 2009)
Management Systems and Credit Standing	1	Yang and Wu (2008)
Manufacturing performance	1	Vachon and Klassen (2006)
Material Quality	1	Thangchattu and Siripokapiram (2010)
Material price	1	Thangchattu and Siripokapiram (2010)
Net life cycle cost	1	Noci (1997)
Operation management	1	Hong-jun and Bin (2010)
Operational capacity	1	Yan (2009)
Organization	2	Buyukozkan and Cifci (2010, 2011)
Organization structure and manufacturing capability	1	Kuo and Lin (2011)
Organizational culture & strategy issues	1	Wen and Chi (2010)
Organizational performance dimension	2	Buyukozkan and Cifci (2010, 2011)
Outside environmental management	1	Hong-jun and Bin (2010)
Packaging/reverse logistics	1	Handfield et al. (2002)
Partnership with green organization	1	Awasthi et al. (2010)
Pollution control	1	Lee et al. (2009)
Pollution control initiatives	1	Awasthi et al. (2010)
Price	1	Yang and Wu (2008)
Price	3	Yang and Wu (2008); Grisi et al. (2010); Yan (2009)
Process management	2	Hsu and Hu (2007, 2009)
Procurement management	2	Hsu and Hu (2007, 2009)
Product attributes	1	Handfield et al. (2002)
Product Cost	1	Feyzioglu and Buyukozkan (2010)
Product Quality	1	Feyzioglu and Buyukozkan (2010)
Profitability attributes	1	Bai and Sarkis (2010a; b)
Profitability of supplier	1	Chen et al. (2010)
Purchasing performance	1	Large and Thomsen (2011)
Purchasing's environmental capabilities	1	Large and Thomsen (2011)
Quality	8	Kuo et al. (2010); Lee et al. (2009); Yang and Wu (2008); Li and Zhao (2009); Yan (2009); Grisi et al. (2010); Zhang et al., (2003); Wen and Chi (2010)
Quality of Delivery	1	Grisi et al. (2010)

Appendix 9 (continued)

Criteria	No. of articles	Author
Quality system	1	Kuo and Lin (2011)
R&D capability	1	Chen et al. (2010)
R&D green products	1	Chen et al. (2010)
R&D management	2	Hsu and Hu (2007, 2009)
Recycle	1	Bala et al. (2008)
Reuse	1	Bala et al. (2008)
Risk factor	1	Chiou et al. (2008)
Service	7	Kuo et al. (2010); Yang and Wu (2008); Wen and Chi (2010); Li and Zhao (2009); Bala et al. (2008); Yan (2009); Feyzioglu and Buyukozkan (2010)
Social responsibility & Environmental competencies	2	Buyukozkan and Cifci (2010, 2011)
Staff Training	1	Awasthi et al. (2010)
Strategic level of purchasing	1	Large and Thomsen (2011)
Supplier relationship	1	Chen et al. (2010)
Supplier's implementation capability	1	Kuo and Lin (2011)
Supplier's green image	2	Wen and Chi (2010); Noci (1997)
Supply chain base	1	Chiou et al. (2008)
Supply chain management	1	Hong-jun and Bin (2010)
Supply chain stages	1	Lu et al. (2007)
Technology	5	Buyukozkan and Cifci (2010, 2011); Li and Zhao (2009); Lee et al. (2009); Wen and Chi (2010)
Total product life cycle cost	1	Lee et al. (2009)
Transportation time	1	Yeh and Chuang (2011)
Us of environmental friendly material	1	Awasthi et al. (2010)
Use environment friendly technology	1	Awasthi et al. (2010)
Waste management	1	Handfield et al. (2002)
Yield	1	Yeh and Chuang (2011)

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